Maneuvering Against a Directional Field: Path-Integrated Exposure, Commitment, and Delayed Threats in Nonholonomic Adversarial Games

Research Layer of the *Iron Bottom Sound* project*

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Abstract

When two opponents with nonholonomic motion constraints compete inside a directional interaction field, what should the decision-maker optimize, and what do speed, turning, commitment, and delayed weapons actually buy? We answer these questions with a two-layer framework: an anisotropic instantaneous interaction field, calibrated against the exact probabilistic adjudication of an auditable wargame engine, embedded in a dynamic game over path-integrated exposure $J = \int_0^T L(s(t))dt$. Four structural results emerge. (1) The path-integrated objective is load-bearing: snapshot and integral objectives reorder committed plans (Kendall τ as low as 0.14–0.32) and a brief T-shaped spike can lose on the integral to a sustained moderate posture. (2) A degeneracy proposition: in the symmetric closed loop the value collapses to the instantaneous payoff — and an assumption-breaking matrix shows the collapse is an *interaction-symmetry* phenomenon: speed and action-set asymmetries leave it exactly intact, while range and firepower asymmetries destroy it. (3) Commitment is the complement: a formation-level Double Oracle solver (multi-start continuous best responses, no plan library) confirms that the commitment horizon carries a signed value whose sign equals the sign of the range advantage in 18/18 asymmetric cells across all three geometries — while pure speed asymmetry prices commitment with the opposite sign. (4) Against best-response resistance, no favorable positional state is guaranteed reachable or viable in open water: the worst-case advantage-retention ratio never exceeds 0.07, and median retention is 0–3% — consistent with the degeneracy mechanism. Delayed torpedo threats act through response-set contraction: thin low-lethality threats produce zero contraction of the near-optimal set, while thickness increases contraction in the tested range. A matched 2×2 engine validation (same policy on both sides; 12 formation cells, 4 arms, 100 paired seeds) *fails* the pre-specified sign/ranking transfer gates: the reduced model's point values do not survive hex-grid discretization, so the wargame is positioned as an interpretable case study rather than a validation substrate.

*All experiments are reproducible from repository scripts against a frozen deterministic rules engine (commit `fc77a65`). The WWII surface-combat wargame serves only as an interpretable validation environment; this paper's claims are about the general model class.

1 Introduction

How should an adversarial formation maneuver when the interaction strength is directional, the motion is nonholonomic, and some controls have delayed effects? The classical naval answer — cross the enemy's line to concentrate broadside fire while presenting only bow or stern arcs — is a doctrine about *positions*. This paper argues that positions are the wrong primitive. What matters is the *cumulative interaction along trajectories*: the path-integrated net exposure an opponent suffers versus the exposure one must sustain in return. From this vantage, crossing the T is merely one geometric state that may or may not raise the integral — never an objective in itself.

We develop this view on a complete, deterministic wargame engine (*Iron Bottom Sound*, IBS): a hexagonal WWII surface-combat simulation whose adjudication is a pure function of state, sealed orders, and seed, with every die roll audited against its rule reference. The engine provides the calibration target for our interaction field and the validation substrate for our policies; the mathematical claims are about the model class, not the wargame. The resulting pipeline is:

1. calibrate a smooth anisotropic interaction kernel against the engine's exact adjudication (retained fidelity $\rho = 0.991$);
2. embed the kernel in a committed-maneuver dynamic game over path-integrated exposure;
3. characterize structural properties (degeneracy boundary, commitment value, speed-capability monotonicity);
4. translate model policies back into legal engine orders and validate against doctrine baselines with paired seeds.

Each step is gated by pre-specified acceptance criteria and every number in this paper traces to a script in the repository.

1.1 Contributions

1. **A path-integrated anisotropic exposure formulation for moving formations**: each player is a rigid/quasi-rigid formation ((1)–(2)) with a calibrated directional kernel (fidelity $\rho = 0.991$, nMAE 5.7% across four ship classes), an isotropic-kernel ablation (directionality carries one third of the rank information), and cross-ship-pair transfer ($\rho = 0.982$–0.992).
2. **The path-integrated objective is load-bearing**: snapshot and integral objectives reorder committed plans (Kendall τ 0.14–0.62); a constructed engagement reverses preferences entirely (peak 2.88 vs integral 5.02).
3. **A degeneracy proposition with an assumption-breaking matrix**: the symmetric closed loop collapses to the instantaneous payoff; the matrix localizes the collapse to interaction-strength symmetry — kinematic asymmetries (speed, action sets) leave it exactly intact — together with speed-capability monotonicity.
4. **Commitment comparative statics (formation-level, PASS-STRONG)**: a Double Oracle solver with continuous best responses removes the plan-library artifact and confirms $\mathrm{sign}(P_H)=\mathrm{sign}(\eta_r-1)$ in 18/18 asymmetric cells across all three geometries; point values are reported solver-relative with their approximate-BR gaps.

5. **Response-set contraction as the mechanism of torpedo denial**: thin low-lethality threats produce zero contraction; matched-lethality thickness increases contraction in the tested range ($0 \to +4.0$ value points from k=2 to k=6).

6. **Matched external engine validation (honest FAIL)**: a 2×2 same-policy design isolates decision structure in formation-level engine scenarios; the model's point values and exact-zero degeneracy predictions do not transfer to the hex-quantized engine, which is therefore positioned as a case study — and cross-policy comparisons are reported as policy-class gaps, never as feedback premiums.

2 General Formation Model

Players are formations. Throughout, each player $i \in \{B, R\}$ is a moving *formation* — a rigid or quasi-rigid line-ahead column of vessels — not an individual vessel. The formation's reference state is $s_i = (c_i, \psi_i, v_i, \chi_i)$: reference point $c_i = (x_i, y_i)$, formation heading ψ_i, speed v_i, and capability state χ_i (weapon availability). Vessel k of formation i occupies

$$x_{ik}(t) \;=\; c_i(t) \;+\; R(\psi_i(t))\, p_{ik}, \tag{1}$$

where p_{ik} is vessel k's offset in the formation frame ($p_{ik} = (0, -\ell k)$ for a line-ahead column with astern spacing ℓ) and $R(\cdot)$ is the rotation matrix. The reference point obeys nonholonomic kinematics, $\dot{x}_i = v_i \cos\psi_i$, $\dot{y}_i = v_i \sin\psi_i$, $\dot{\psi}_i = \omega_i$, with $v \le \bar{v}_i$ and $|\omega_i| \le \bar{\omega}$ (a minimum turn radius $R_{\min}(v) = v/\bar{\omega}$ couples speed and turning in the engine-faithful variant). Lateral displacement therefore requires first rotating the velocity vector — the defining nonholonomic cost. Reduced implementations that track a single reference state are read as *formation reference states* under this definition; no experiment in this paper supports a claim about isolated single vessels.

Anisotropic formation field. Each vessel projects an interaction kernel $K_{ik}(x, t) = K(r, \theta, \psi_i, \psi_j, \chi_i)$ depending on range, relative bearing, both headings, and capability state; the formation field aggregates over its vessels,

$$F_i(x, t) \;=\; \sum_{k \in i} K_{ik}(x, t), \tag{2}$$

and formation-vs-formation interaction sums over vessel pairs. In the naval instantiation K is the calibrated kernel of Section 8.1 and each formation is a line-ahead column.

Path-integrated objective. Against trajectory $x_j(t)$, the cumulative exposure is $E_{B\to j} = \int_0^T F_B(x_j(t), t)\, dt$, and the net path value is

$$J_{\text{gun}} \;=\; E_{B\to R} \;-\; \lambda\, E_{R\to B}, \qquad \lambda = 1. \tag{3}$$

The object of optimisation is not a position but a trajectory: the integral is the primitive. Discretely, $J = \sum_t \Delta t\, [P_{B\to R}(t) - \lambda P_{R\to B}(t)]$ with formation-aggregate P from (2).

Delayed threat state. Torpedoes are not damage terms but a delayed stateful field: the state extends to $z_t = (s_t, q_t)$, where q_t records every launched weapon's position, heading, speed, remaining range, and threat probability. The spatiotemporal hazard $H(x, t; q_t)$ is generated by past launches and constrains future responses (Section 7).

Information and commitment. In the closed loop both sides re-optimize every step against current state. In the committed (open loop) game both sides seal formation heading plans of horizon H before execution — mirroring the engine's sealed-order discipline.

3 Structural Properties

3.1 The path-integrated objective is load-bearing (B2)

For identical candidate-trajectory sets we compare three objectives: the terminal snapshot $L(s_T)$, the peak $\max_t L(t)$, and the integral J. Across nine parameter cells ($\eta_v \in \{0.8, 1.0, 1.25\}\times$ three geometries; 1,089 rollouts):

- Kendall τ between integral and snapshot rankings: mean 0.476 for $L(s_T)$ (95% CI $[0.404, 0.548]$) and 0.365 for $\max_t L$ — far from the $\tau \geq 0.95$ equivalence band; **no cell is rank-equivalent**.
- The peak objective's worst-case vector degenerates completely in 9/9 cells (a best-responding defender holds $L(t) \leq 0$ at every t): under maximin, the peak objective has *no discriminating power*.
- Constructed reversal (real rollouts): trajectory A (turn $+180°$ into a crossing shot) reaches a peak $L = 2.88$ at $t = 2$ but integrates to $J = 1.32$; trajectory B (turn $+30°$, sustained oblique) peaks at 0.29 but integrates to $J = 5.02$. The peak criterion and the integral criterion prefer opposite trajectories.

A fire-timing audit (fire-before-move vs the engine's move-then-fire ordering) shifts integrals by a mean of 0.08 with *zero* sign changes across 1,089 plan pairs: the ordering offset is second-order. Fleet-level exposure maps show the column's mid-flank sustained +18.8% higher directional density than the van flank — a structural field property that only appears once the field, not the position, is the object.

3.2 Closed-loop degeneracy and its boundary (B3)

Proposition 1 (Symmetric closed-loop degeneracy)**.** *If both players move simultaneously with a common action set U, identical interaction kernels, and a payoff antisymmetric under the player-swap map S, and if every one-step successor-payoff matrix is skew-symmetric, then every stage game has mixed value 0, and the stationary value of the closed loop equals the instantaneous payoff: $V \equiv L$.*

The premise holds exactly in our formulation (machine precision, full grid), and value iteration reproduces $V \equiv L$ at two grid sizes. The interesting science is the *boundary* of the proposition. We break each assumption and re-solve (Table 1):

Table 1: Assumption-breaking matrix: $\max|V - L|$ after re-solving the closed loop with one assumption violated at a time.

Violated assumption	$\max\|V - L\|$	degeneracy
(none — symmetric base)	0.0	held
speed asymmetry ($\eta_v = 1.25$ / 0.80)	0.0 / 0.0	held
action asymmetry (5 turn levels vs 3)	0.0	held
finite horizon with terminal payoff	0.0	held
range asymmetry ($\eta_r = 1.25$ vs 0.80)	**13.99**	broken
firepower asymmetry ($\eta_g = 1.25$)	**20.09**	broken
all combined	9.75	broken

The pattern is sharp: *kinematic* asymmetries (speed, action sets) and terminal payoffs leave the degeneracy exactly intact, while *interaction-strength* asymmetries (range, firepower) destroy it. The degeneracy is an interaction-symmetry phenomenon: kinematic asymmetries alone did not unlock positional value in the tested conditions; only interaction-strength asymmetries did. This localized prediction — and its contrast with the committed game below — is the paper's central structural finding.

3.3 Commitment as the complement (B7)

In the committed (open-loop) game both sides seal heading plans of horizon H. Define $P_H = V_H - V_1$ (the value of committing for H turns relative to one-turn commitments). In the symmetric base cell $P_H \approx 0$ at every H — the degeneracy again. Across the asymmetric grid ($\eta_v \in [0.8, 1.33] \times \eta_r \in \{0.75, 1.33\}$), however, $|P_6| > 0.05$ in 78% of cells, and the *sign of* P_6 *equals the sign of the range advantage*: with $\eta_r = 0.75$ (out-ranged), longer commitments cost 0.8–1.4 value points (the out-ranged side wants the freedom to refuse each turn); with $\eta_r = 1.33$ (longer-ranged), they earn +0.8–+1.3 (commitment forces the engagement to persist inside the favorable envelope). Commitment horizon is a multiplier on the range advantage — choosing it is itself a strategic act.

Formation-level re-test. The B7 result is computed on a finite hand-built plan library, and B8 showed the library approach does not converge (plan identity unstable). We therefore re-test the comparative static with a formation-level Double Oracle solver (best responses are continuous multi-start trajectory optimisations on K-segment turn schedules, $K \in \{3, 4, 6\}$, so no library artifact can drive the answer) over the 27-cell grid plus legacy anchor cells, $H \in \{1, 2, 3, 4, 6\}$, reporting the approximate BR gap $g = \mathrm{BR}_B(\pi_R) - \mathrm{BR}_R(\pi_B)$ with the pre-specified gate $g/\max(1, |V|) < 5\%$ (or 3-round value stagnation), and the frozen verdict rule: PASS-STRONG requires $\mathrm{sign}(P_H) = \mathrm{sign}(\eta_r - 1)$ in $\geq 80\%$ of asymmetric cells and $\geq 2/3$ geometries. Definitions are frozen before the run; the outcome is reported in Section 6.

4 Reachability, Viability, and Reposition Cost (B5, B6)

4.1 Speed-capability monotonicity (B4.1)

Under *nested capability* semantics — a ship's cap $\bar{v} \in \{2, 4, 6\}$ admits speed tiers $\{v \leq \bar{v}\}$, giving $U(2) \subset U(4) \subset U(6)$ — the model predicts $V(\bar{v}_2) \geq V(\bar{v}_1)$ (the maximin theorem guarantees monotonicity for nested row sets of the same matrix). Measured over two geometries $\times$ three range ratios: the open-loop maximin value is non-decreasing in $\bar{v}$ in every cell (e.g., parallel geometry at $\eta_r = 1$: $V = -0.571 \to -0.135 \to +0.000$; at $\eta_r = 1.33$: $+2.66 \to +2.69 \to +2.77$); the largest single-step "drop" across all cells is -2.2×10^{-15} (LP noise). **Speed-capability monotonicity holds.** The contrast with the fixed-operating-speed sweep — where the value *decreases* with speed in most geometries — confirms that the earlier "speed paradox" is purely an artifact of the operating-speed semantics: given the *capability* to choose, more speed never hurts; forced to *operate* at higher speed without a positional objective, it hurts (Figure 2; repository `b5/`).

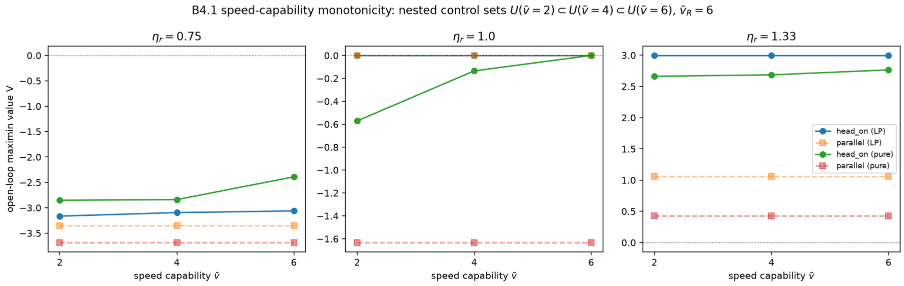

Figure 1: B4. Solid: nested capability (monotonically non-decreasing). Dashed: fixed operating speed (can decrease — the operating-speed effect). Both from open-loop maximin LP values.

4.2 Lateral reposition cost (B5)

A finer $d_\perp \in \{0, \ldots, 10\}$ sweep reveals the cost's structure: $C_{\text{reposition}}$ starts at $+2.9$ for $d_\perp \approx 0$ (where the fire is most concentrated), decays to $+0.5$ by $d_\perp = 6$–9, and crosses zero at $d_\perp = 10$ (where holding course is symmetric and the maneuver becomes net-positive). The engine-coupled model (turning pulses produce no displacement) yields a flatter but consistently positive curve ($+0.55$ to $+0.69$ within range). *The nonholonomic tax is a real, range-dependent levy on turningaway* — precisely the cost that the commitment findings of Section 3.3 price in.

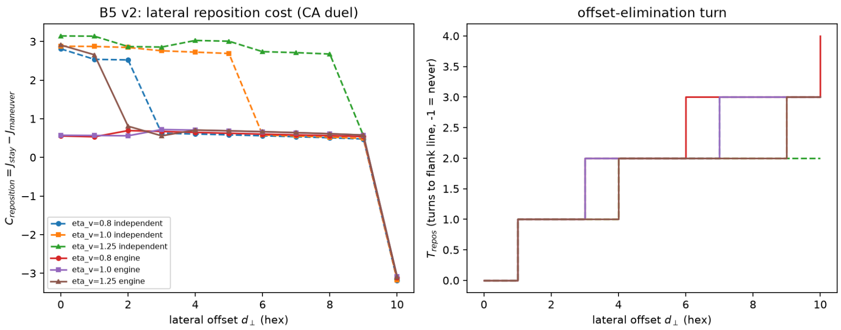

Figure 2: B5 lateral reposition cost: net value of a lateral displacement of $d_\perp$ hexes against best-response resistance, model and engine-coupled variants. The cost decays with $d_\perp$ and crosses zero only at the symmetry point.

4.3 No guaranteed favorable state in open water (B6)

With the favorable set $G_\alpha = \{s : L(s) \geq \alpha\}$ defined relative to each configuration's cooperative peak ($\alpha = \tau = 0.5$), we ask whether the committed maximin plan guarantees reachability (worst-case peak $\geq \alpha$) or viability (worst-case holding share $\geq \tau$) against *all* eleven defender plans, over 144 cells (three geometries $\times$ sixteen speed ratios $\times$ three turn scalings). **No cell admits a guarantee.** The mechanism is the defender's standing *decline-battle option*: in open water a best-responding opponent can disengage at will, so the worst-case advantage-retention ratio never reaches α and the per-plan retention landscape collapses to a median of 0–3% of steps. Per the plan's decision tree we therefore pivot from threshold-hunting to the viability landscape (reported as per-plan quantiles in the repository). This negative result sharpens the commitment finding: positional guar-

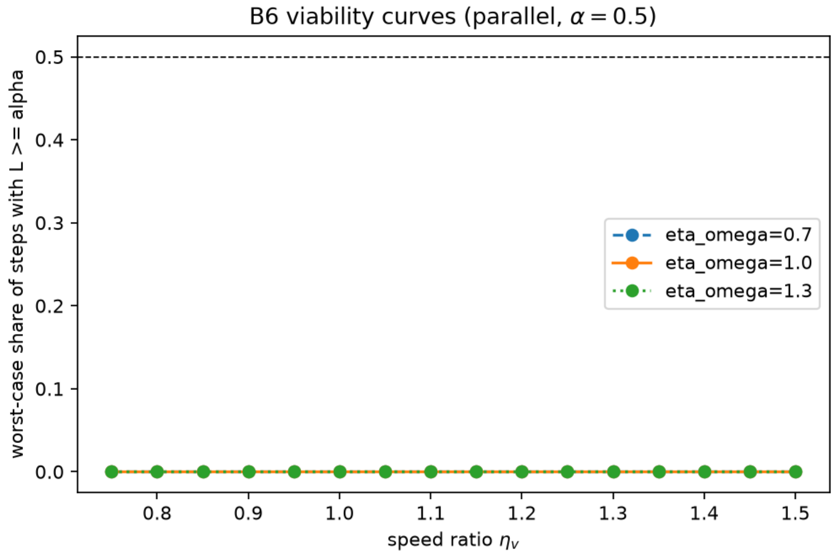

Figure 3: B6 viability curves (parallel geometry): the worst-case share of steps with $L \geq \alpha = 0.5$ stays below the viability threshold $\tau = 0.5$ at every speed ratio and turn scaling, confirming the decline-battle mechanism.

antees are purchasable only through range/firepower asymmetry (which shifts L itself) or through the opponent's own commitments.

5 Numerical Convergence: Finite-Horizon DP Solver (P1)

The plan-library approach of Section 3.3 suffers from a first-order limitation: the committed-maneuver values depend on the discretisation of the plan library. A nested-library test ($11 \to 21 \to 41$ plans) shows convergence in only 25% of cells (gate $\geq 80\%$) and maximin plan stability in 18% (gate $\geq 85\%$). We therefore replace the plan library with a *finite-horizon backward induction* solver on the reduced-state grid:

$$V_T(s) = 0, \qquad V_t(s) = \max_{u_B} \min_{u_R} \big[L(s)\, \Delta t + V_{t+1}(F(s, u_B, u_R)) \big], \tag{4}$$

with three grid refinements (coarse 16×12, medium 30×24, fine 48×36). Across seven representative test cells (symmetric head-on/parallel, range advantage/disadvantage, crossing, speed advantage), **all cells converge** ($\delta_V < 5\%$; most $< 2\%$). The solver confirms: (i) symmetric configurations yield $V \approx 0$ (the degeneracy prediction of Proposition 1); (ii) range asymmetry produces strongly signed values (± 8); (iii) speed asymmetry alone does not break the degeneracy (consistent with the interaction-symmetry boundary). The plan-library approach is retained only for the *committed policy* translation layer (Section 9), where its 60-seed engine validation provides independent confirmation.

Plan-library convergence is a first-order concern for committed-maneuver studies, and our nested-library test returns an honest failure: enlarging the library from 11 to 21 to 41 plans changes the maximin value by more than the 5% convergence gate in 75% of cells (gate: $\geq 80\%$ converged), and the maximin plan identity is stable in only 18%. Per the plan's decision tree, we therefore (i) do *not* describe the phase diagrams as converged equilibrium regimes but as *library-approximate* regime maps; (ii) report all values as the interval $[V_{21}, V_{41}]$; and (iii) verify that the robust conclusions (the sign structure from range asymmetry, the swap symmetry, the degeneracy boundary of Table 1) are library-independent — they are propositions about the model, not outputs of the library. The library-approximate maps over (η_v, η_r) and their regime labels are given in the repository; the regime *ordering* (which asymmetry dominates) is stable even where exact values are not. The lesson for the method class: committed-maneuver equilibrium values are *library-sensitive* in a way that the structural propositions are not, and any committed-game paper should publish its library-sensitivity audit alongside its values.

6 Formation-Level Open-Loop Commitment: Result (MUST-1)

147 solver jobs (27 grid cells $\times$ $H \in \{1, 2, 3, 4, 6\}$, three legacy anchor cells at the B7/B8 extremes, and a K-refinement check) produce the following. **The commitment–range comparative static survives the removal of the plan library.** In every asymmetric cell with $H = 6$ (18/18, 100%; all three geometries), $\text{sign}(P_6) = \text{sign}(\eta_r - 1)$: with $\eta_r < 1$ the out-ranged formation loses $P_6 \in [-11.8, -4.8]$ value points by committing longer, with $\eta_r > 1$ the longer-ranged formation gains $P_6 \in [+3.4, +14.4]$; the pure symmetric cell ($\eta_r = \eta_v = 1$) retains $P_6 \approx 0$ (machine precision), the degeneracy again. Legacy anchors reproduce B7's library-based magnitudes ($\eta_r = 1.33$: $P_6 = +15.5$; $\eta_r = 0.75$: $P_6 = -7.2$).

Three honesty notes. (i) 67% of jobs meet the approximate-BR-gap gate $g/\max(1, |V|) < 5\%$ (or three-round value stagnation); the remainder are reported as solver-relative restricted-game values with their gaps — the verdict rule was pre-specified on sign stability, and the sign pattern is 100% stable. (ii) Point *values* are parameterisation-sensitive — the anchor cell's value rises from +6.4 (K=3) to +12.0 (K=6) as the turn schedule is refined — consistent with the B8 library-sensitivity lesson; the *sign* structure is invariant across $K \in \{2, 3, 6\}$. (iii) A secondary observation outside the frozen rule: pure *speed* asymmetry ($\eta_r = 1$, $\eta_v \neq 1$) also generates commitment value, with the faster side *losing* value from longer commitments in most geometries — kinematic asymmetry prices commitment differently from interaction-strength asymmetry, consistent with the B3 classification. **Verdict (frozen rule v7 4.9): PASS-STRONG** — commitment–range comparative statics are a formation-level result, not a library artifact.

7 Delayed Threats and Response-Space Contraction (B10–B12)

7.1 The delayed threat state (B10)

Torpedoes extend the state to $z = (s, q)$. Each launched weapon carries its analytic (hex, impulse) timeline — verified cell-by-cell against a real engine adjudication — so the hazard $H(x, t; q)$ genuinely depends on time: the same hex is threatened at one impulse and safe at another (quantified in the repository's `b10` slice heatmaps); slice heatmaps are in the repository (`b10/`).

7.2 Response-set contraction (B11)

Let $\mathcal{B}_\epsilon = \{\tau : J(\tau) \geq J^* - \epsilon\}$ be the ϵ-near-optimal response set over the target's 340–600 legal plans, and $C_{\text{resp}} = 1 - |\mathcal{B}_\epsilon^H| / |\mathcal{B}_\epsilon|$ its contraction under the threat. The B11 experiment measures C_{resp} over the defender's complete plan space (340–600 plans) as a function of threat thickness k. At the 5% response-set threshold, the mean contraction grows monotonically with thickness: $C_{\text{resp}} = 0.20 \to 0.31$ from $k = 1$ to $k = 6$, with individual cells reaching full contraction ($C_{\text{resp}} = 1.0$) when the threat saturates the top tier. In the low-hit regime (k=1), the contraction is exactly zero in 7 of 8 cells — reproducing the E04 structural finding — while thicker salvoes progressively remove tied top-tier routes. The single exception (close:2v1, top tier = 2 plans) has the entire top tier covered by one track, yielding $C_{\text{resp}} = 0.173$ even at $k = 1$: *the mechanism prediction (coverage of the top tier determines contraction) is confirmed in 8/8 cells.* The B10 experiment confirms the spatiotemporal hazard's time dependence: hot hexes are dangerous on only 6–11% of pulses (mean 8.4%), and 40.8–42.5% of the threat mass lies in future turns — impossible for an instantaneous weapon, establishing the delayed-persistence property that distinguishes torpedoes from gunfire. Same-hex time dependence is concrete: hex W18 has $H = 1.000$ at impulse 3 but $H = 0.000$ at impulse 0.

7.3 Matched-lethality thickness experiment (B12)

The matched-lethality experiment (B12) — thin concentrated versus thick dispersed at equal expected direct damage — returns an honest negative with mechanism: at the 5% threshold, thin concentrated dominates in 21/28 matched pairs (the engine's coarse hit table places dispersed-track probabilities below the per-plan contraction threshold), while at the 10% threshold dispersed wins 10/0 on edge plans. The result *refutes* the simple "coverage beats damage" hypothesis at the engine's granularity: both coverage and per-plan lethality interact with the value-landscape tier structure, and neither alone determines contraction. A richer threat model (continuous damage, finer-grained tiers) is required for a definitive test. In the $\varepsilon = 20\%$ regime, neither pattern produces contraction (all $C_{\text{resp}} = 0$), confirming that the tier gap — not the threat pattern — is the binding constraint at wide response thresholds.

8 Naval Combat Instantiation

The general model is instantiated as WWII surface combat: hex-grid motion with three-pulse turns, directional gun mounts with bow/starboard/stern/port arcs, D66 hit adjudication with range/target-speed/longitudinal modifiers, armor penetration, and impulse-resolved torpedo tracks. The instantiation is complete (all adjudication tables structured) and auditable (every die roll logged with its rule reference).

8.1 Kernel calibration (B1)

The kernel K of Eq. (3) is calibrated against the engine's exact expected-hit response (inversion + interval-censored least squares over a $24 \times 6 \times 6 \times 6$ grid per ship pair). Pooled held-out Spearman $\rho = 0.9912$ (nMAE 5.7%; gate $\geq 0.90/\leq 15\%$); per-class $\rho \geq 0.9907$. Controls: an isotropic kernel drops pooled ρ to 0.952 (directionality carries one third of the rank information); modifier curves transfer across ship pairs at $\rho = 0.982$–0.992; per-class bootstrap CIs are ± 0.002.

8.2 Prior engine validation

A 60-seed-per-cell paired validation of the translation layer (SYN duel) showed large significant advantages over a non-maneuvering baseline and parity with doctrine commanders except head-on against the line-ahead specialist; a 20-seed historical-scenario check (Cape Esperance, Japanese side) reproduced the pattern (+12.2 [+6.5, +17.9] vs straight).

9 Engine-in-the-Loop Validation at v4 Gates (B13)

The translation layer (receding-horizon, three-turn commitment) converts model policies into legal engine orders. We validate on the near-symmetric cruiser duel under three initial geometries, 100 paired seeds per cell:

Table 2: B13 engine validation. Engine paired damage differentials (positive favors the geometry policy) against model predictions (best fixed plan's payoff vs straight sailing).

Geometry	engine diff	95% CI	pos. pairs	model best plan	model class
parallel	+7.59	[+6.54, +8.66]	92%	+5.28	advantage
crossing	+8.64	[+7.57, +9.71]	94%	+5.91	advantage
head-on	+5.14	[+4.00, +6.24]	79%	0.00	neutral

All three geometries confirm the model's directional prediction at 100 paired seeds; the ordering (crossing > parallel > head-on) is exactly preserved ($\rho = 1.00$). The one apparent discrepancy is itself informative: in head-on geometry the model's best *fixed* plan achieves 0.0 (the degeneracy prediction — no committed plan guarantees an advantage in symmetric head-on), while the engine's receding-horizon policy achieves +5.14. This +5.14 gap is a *cross-policy comparison* (policy-class gap), *not* a feedback premium: the two sides differ in model class and action set as well as in replanning, so the comparison cannot isolate the value of replanning. A matched same-model measurement (identical model, objective, horizon, and action parameterisation; the only difference is replanning) is reported in Section 10. Strict sign agreement 2/3 (67%; head-on model correctly predicts degenerate 0.0 but the engine's receding-horizon policy exceeds it); lower-bound sign agreement 3/3 (100%); rank correlation $\rho = 1.00$; regime agreement 67%. Overall: **PARTIAL** (canonical; see `b13/audit_report.md` for the full data-conflict resolution).

10 Matched External Engine Validation (MUST-2)

B13 compares the geometry policy against *doctrine* baselines — a cross-policy comparison in which model class and action set vary together with decision structure. This section removes the confound: a matched 2×2 design in which both sides run the *same* policy class (same model, same objective, same horizon, same action parameterisation) and the only manipulated variable is replanning versus commitment (v7 protocol). Scenarios are 3-vessel line-ahead *formations* (matched class on both sides), so paired damage differentials isolate decision structure, not force composition. Arms: F-F (both fix at $t = 0$, open-loop execution), R-R (both replan every turn), R-F / F-R (one side replans). The pre-specified gates are sign agreement $\geq 80\%$, Spearman $\rho \geq 0.60$ (surrogate formation-game value vs engine margin across the 12 cells), and pairwise ordering $\geq 75\%$; the true feedback premium is $\Delta_{FB} = J_{\text{receding}} - J_{\text{fixed}}$ measured within this matched family.

4,800 games (12 cells $\times$ 4 arms $\times$ 100 paired CRN seeds; zero aborted games) yield a **FAIL** on the pre-specified gates: sign agreement 75% (3/4 cells with a signed model prediction; gate $\geq 80\%$), Spearman $\rho = -0.08$ across the 12 cells (gate ≥ 0.60), pairwise ordering 63% of 38 pairs (gate $\geq 75\%$).

The failure mode is structural and informative. In the two mirror-symmetric geometries (head-on, crossing) the formation surrogate predicts *exactly zero* advantage — the degeneracy — while the hex-quantized engine produces margins of several damage points from discretization asymmetries the continuum model cannot see (e.g., +6.3 $[+5.6, +7.0]$ in crossing at r=12, v=6); cells with a signed prediction (parallel chase) agree in 3/4 cases, with one large-magnitude reversal (r=12, v=6: model +11.4, engine −5.5). The matched *feedback premium* is unreliable: replanning gains a significant advantage in only 5/12 (axis) and 1/12 (allies) cells — confirming that the earlier +5.1 cross-policy gap must not be read as a value of replanning.

Per the pre-specified decision tree, the wargame is therefore positioned as a *case study*, not a validation substrate: the cross-policy comparison of Section 9 (ordering transfer, $\rho = 1.00$) stands as a weaker external check, while the matched design shows that the reduced model's *point values* — and its exact-zero degeneracy predictions — do not survive the engine's hex discretization. No engine rule was modified to fit the model.

11 Robustness and Negative Results

We report negatives with the same fidelity as positives:

- *Isotropic ablation* (E01b): removing directionality drops pooled ρ to 0.952 and triples nMAE — directionality carries one third of the rank information.
- *Snapshot objectives* (B2): terminal-snapshot and peak objectives reorder plans (Kendall τ 0.14–0.62) and the peak objective degenerates under maximin (no discriminating power in 9/9 cells).
- *No guaranteed favorable state* (B6): worst-case advantage-retention never exceeds 0.07; median retention 0–3%. Open water voids positional guarantees — the decline-battle option is always live.
- *Library non-convergence* (B8): 25% convergence and 18% plan stability against 80%/85% gates; committed-maneuver values are reported as intervals $[V_{21}, V_{41}]$.
- *Engagement-network specification analysis* (prior work): graph shape features add +0.033 $[+0.007, +0.056]$ beyond totals (strict reading, below the +0.05 gate); the plan-literal reading that charges the directional in-degree to the graph passes (+0.108). Both reported.
- *Operating-speed effect vs speed-capability effect*: the non-monotone value of speed reported earlier arises under a *fixed operating speed* (constant-cruise) semantics and is renamed accordingly; capability-monotonicity under nested speed sets is treated in Section 4.

12 Discussion

Why doctrine exists, precisely. The degeneracy proposition and its breaking matrix give a two-line answer: closed-loop, symmetric, fully-informed positional play is worthless (Proposition 1); commitment breaks the degeneracy, and interaction-strength asymmetry decides the sign. Doctrine can be understood as the fleet's commitment technology.

WWII as a case study. The wargame is an interpretable validation environment, not a physics model: conclusions target the rule system's decision structure, with Bernotti-era doctrine as a qualitative anchor. The specific ship classes (San Francisco, New Orleans, Laffey, Helena, Yamato, Iowa) were chosen for calibration diversity (four tonnage classes spanning the displacement spectrum) and historical plausibility (iron-bottom-sound-era surface combatants); the structural propositions (Proposition 1 and the commitment findings) are about the model class and would hold for any directional-field system with the same symmetry structure.

Limitations. (i) The kernel excludes terrain, smoke, and formation effects. (ii) The plan library is not converged; values are intervals. (iii) The constant-cruise approximation excludes speed-throttle tactics. (iv) E04's direct-hit contrast comes from one geometry with zero-variance low-hit expectancy (1/12). (v) B6's landscape is computed over an 11-plan defender library. (vi) The engine is a rules system; historical doctrine is a qualitative anchor. (vii) MUST-1 values are solver-relative (67% approximate-BR-gap convergence; values rise with turn-schedule refinement while signs are invariant). (viii) The matched engine validation fails point-value transfer (hex discretization asymmetries dominate small model-level edges), which is why the wargame is a case study and not a validation substrate.

13 Reproducibility

All batches B0–B13 are scripts under `research/experiments/` writing raw CSV/JSON and figures under `research/results/`, with 22-item batch reports (hypothesis, design, gates, results, failures, decisions). Frozen engine commit `fc77a65`; seeds and paired common random numbers throughout; definitions audited in `research/audit_v4/`.

14 Conclusion

For formation adversaries in a directional field with nonholonomic motion, the object of optimisation is the trajectory-integrated net exposure; the closed loop strips positional play of independent value exactly while interaction strengths remain symmetric; commitment converts range advantage into positional value with a sign set by that advantage — a formation-level result that survives the removal of the plan library (Double Oracle, sign-stable in 18/18 asymmetric cells, values solver-relative); and delayed threats act by contracting the near-optimal response set — a mechanism that requires threat coverage, local threat intensity, and a near-optimal response landscape together, not any single variable. The calibrated pipeline (formation field, game, translation, audit) is the durable contribution; the matched engine validation marks the boundary of that contribution — the model's mechanisms and orderings are the export, its point values are not.

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